

KrishiSetu AI: A Safe-by-Construction 4-Tier Hybrid Decision and Evidence Architecture Anchored on Agricultural Digital Public Infrastructure

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Abstract—Indian smallholder agriculture faces a persistent last-mile translation bottleneck: although official weather forecasts, digital soil-health records, georeferenced plot surveys, and a growing corpus of agronomic advisories are now available at national scale through Agricultural Digital Public Infrastructure (DPI), the majority of farmers still receive static, context-free alerts that cannot, by themselves, prevent crop failure. National DPI has already produced unprecedented datasets—as of August 2026 the AgriStack Farmer Registry has issued more than 10.33 crore unique Farmer IDs, the Kharif 2025 plot survey has georeferenced 28.5 crore plots, and the Kisan e-Mitra conversational assistant has answered more than 93 lakh farmer queries in eleven regional languages. These assets, however, stop short of plot-specific decision support, and the manual, paper-based evidence pipeline of the Pradhan Mantri Fasal Bima Yojana (PMFBY) still delays loss settlement by months. We present KrishiSetu AI, a safe-by-construction, four-tier hybrid decision and evidence architecture that anchors on this existing DPI and converts government and remote-sensing data into concrete weekly field actions. Tier 1 is a deterministic rule engine immune to hallucination; Tier 2 is a deterministic canonical-knowledge layer for compliance-critical facts; Tier 3 is a domain-governed retrieval-augmented generation layer; and Tier 4 is an LLM that renders, but never invents, decisions. A composite confidence index Γ , formed from the tier signals, gates every response against a threshold $\tau = 0.85$, escalating uncertain or malicious queries to human KVK agronomists. A Sentinel-2 NDVI temporal-anomaly pipeline produces insurance-ready crop-damage dossiers that complement PMFBY’s existing CROPIC/YES-TECH workflows. In controlled benchmarks the system achieves a 91.4% top-1 accuracy on PlantVillage and 73.2% on the in-the-wild PlantDoc set, a 14.2% structured abstention rate on 500 adversarial queries, and a zero chemical-dosage hallucination rate, with median advisory latency of 3.8 s on commodity hardware.

Index Terms—Digital Public Infrastructure (DPI), Retrieval-Augmented Generation (RAG), agronomic decision support, smallholder farmers, PMFBY, Sentinel-2 NDVI, hybrid decision engine, large language models, IVR telephony.

I. INTRODUCTION

India’s agricultural sector supports roughly 600 million people yet remains acutely vulnerable to information asymmetry at the farm level. The policy response has been a

substantial build-out of Digital Public Infrastructure (DPI): the Digital Agriculture Mission (DAM) federates AgriStack—a unique 12-digit Farmer ID linked to digitised land records, a geo-referenced plot registry, and a crop-sown registry—with the Krishi Decision Support System and the Digital General Crop Estimation Survey [14]. The results are concrete. As of 6 August 2026, more than 10.33 crore Farmer IDs have been issued and 28.5 crore plots were geo-referenced in the Kharif 2025 survey [10]; the Kisan e-Mitra assistant has answered over 93 lakh farmer queries across eleven regional languages [11]. Indian farmers do not lack information. They lack a trusted, plot-specific decision service that turns scattered government and weather datasets into the answer to a single question: “*What do I do on my plot this week?*”

Static advisories such as “rain expected tomorrow” are routinely ignored because they omit the soil, crop-stage, and variety context that makes an alert actionable. We take the view that a farmer does not need a more fluent chatbot; they need a system that is *safe by construction*—one in which no generative step can propagate a hallucinated dosage or a wrong nutrient-timing decision into a destructive field action, because inputs represent 40–60% of cultivation costs for a smallholder. The PMFBY loss-assessment workflow illustrates the stakes: physical inspection by a Circle Officer followed by manual report preparation can extend claim settlement to six to eight months [9]. PMFBY already pilots technology-based assessment through CROPIC and YES-TECH remote-sensing yield estimation [9]; we extend this logic to the plot level with an automated, insurance-ready crop-damage dossier.

This paper makes four contributions.

- **4-Tier Safe-by-Construction Hybrid Decision Engine.** We separate deterministic agronomic rules, canonical knowledge, domain-governed vector RAG, and LLM rendering so that no hallucination from the LLM tier can propagate into a farm action [4], [5]. This separation is the architectural centrepiece of KrishiSetu AI.
- **Composite Confidence Index.** We aggregate a deterministic rule-match score, a retrieval cosine-similarity score,

and an edge-vision softmax probability into a single gate Γ , delivering an advisory only when $\Gamma \geq \tau$ and otherwise escalating to human KVK officers.

- **Sentinel-2 NDVI Temporal-Anomaly Evidence Pipeline.** We quantify plot-level vegetation loss over a disaster window and emit a structured, insurance-ready damage dossier compliant with PMFBY evidence requirements.
- **Controlled Benchmarking.** We report a 91.4% PlantVillage / 73.2% PlantDoc vision accuracy, a 14.2% structured abstention rate over 500 adversarial queries, and a zero chemical-dosage hallucination rate on the ICAR/KVK corpus.

The remainder is organised as follows. Section II reviews related work. Section III details the system architecture and the four-tier decision engine. Section IV formalises the NDVI anomaly detector and the composite confidence index. Section V presents the experimental setup and benchmarks. Section VI covers the field pre-pilot, DPI alignment, and deployment economics. Section VII discusses limitations and ethics, and Section VIII concludes.

II. RELATED WORK

A. Conversational and RAG-based Agricultural AI

Farmer.Chat demonstrated that retrieval-augmented conversational agents can scale agricultural advisory services to smallholder farmers in sub-Saharan Africa and South Asia, achieving broad thematic coverage of crop management, pest identification, and market pricing [1]. AgriLLM showed that grounding LLM inference in curated, authoritative documents substantially reduces domain-specific hallucination for agricultural knowledge transfer [2]. The newer AgriGPT ecosystem extends this to an LLM-centred agriculture platform [3]. Government assistants such as Kisan e-Mitra operate primarily at the advisory layer [11]. What these systems share is an abstention from hard safety guarantees and from live plot-level evidence integration.

B. Human-in-the-Loop and Safety

Human-in-the-loop LLM systems for farm management decision support have shown empirically that architectures in which domain experts review and override model outputs yield higher farmer trust and lower error rates than fully automated pipelines [4]. We operationalise this finding as an automatic escalation gate rather than a manual review step.

C. Remote Sensing for Crop Assessment

India’s CROPIC and YES-TECH programmes demonstrate that Sentinel-2 multispectral imagery can detect crop-area loss at district scale [9]. The PlantDoc dataset benchmarked in-the-wild, multi-class plant disease recognition under realistic field conditions [6], while PlantVillage supplied the controlled baseline against which deep models are calibrated [7], [8]. Table I positions KrishiSetu AI against these streams of work across the axes that we argue matter: plot-action depth, deterministic safety, explicit confidence gating, automated insurance evidence, and multi-channel delivery.

III. SYSTEM ARCHITECTURE AND THE 4-TIER DECISION ENGINE

A. Overall Data Flow

Figure 1 shows the end-to-end pipeline. A farmer interaction—triggered by a scheduled cron event, a WhatsApp message, an IVR call, or a direct web-app request—enters a *Triage Agent* that identifies intent, language, and delivery channel. The *Context Assembler* then gathers every available plot-specific signal: the consented demo Farmer-ID profile, current crop and sowing date, Soil Health Card parameters, the IMD 7-day district forecast, and the latest Sentinel-2 NDVI composite for the geo-referenced plot boundary. This enriched context object is passed through the four decision tiers in strict order, and the final structured response is rendered in the farmer’s language and dispatched over the appropriate channel.

B. Tier 1: Deterministic Rule Engine

The deterministic rule engine is a pure Python module that encodes agronomic rules as first-class code objects rather than natural-language prompts. Rules are organised into three categories.

- **Weather-to-Action Matrices.** Each entry maps a tuple $\langle \text{crop stage, IMD alert type, soil drainage class} \rangle$ to a ranked action list. For example, $\langle \text{SUGARCANE-GRAND-GROWTH, HEAVY-RAIN-WARNING, M} \rangle$ triggers (a) suspend irrigation, (b) clear furrow outlets, and (c) delay foliar spray by 72 h.
- **Spray Windows.** A boolean function $\text{sprayOK}(t, w)$ returns true only when simultaneously wind speed $w < 10$ km/h, no precipitation forecast within 24 h, relative humidity $< 85\%$, and temperature between 18°C and 35°C .
- **Irrigation Scheduling.** Deficit-irrigation triggers are computed from the IMD district evapotranspiration estimate and the farmer-specified soil water-holding capacity, producing a deterministic irrigation-due flag and a recommended volume in mm/ha.

Because Tier-1 rules are pure Python conditionals, they produce zero hallucinations by construction; the LLM is never consulted for any output Tier 1 can resolve definitively.

C. Tier 2: Canonical Knowledge Layer

ICAR Package-of-Practices documents, minimum-support-price schedules, seed-spacing norms, and scheme-eligibility criteria are stored as Markdown files with typed YAML frontmatter carrying the fields `crop`, `state`, `season`, `valid_from`, and `source_authority`. Documents are loaded deterministically at query time through an exact metadata-filter lookup; they are never chunked, embedded, or fed to a vector index, because stable compliance-critical facts must not be subject to approximate nearest-neighbour retrieval. This guarantees that spacing norms and MSP rates are always served from an authoritative source with a known validity date.

TABLE I: Positioning of KrishiSetu AI against prior and contemporary systems for smallholder agricultural AI.

Capability	Farmer.Chat	AgroLLM	Kisan e-Mitra	HITL Farm Mgmt	KrishiSetu AI
Plot-action decision depth	×	×	×	partial	✓
Deterministic safety tier	×	×	×	partial	✓
Explicit confidence gating / abstention	×	×	×	✓	✓
Automated NDVI / PMFBY evidence	×	×	×	×	✓
Multi-channel (web/WhatsApp/IVR)	partial	×	partial	×	✓
Expert (KVK) escalation	×	×	partial	✓	✓

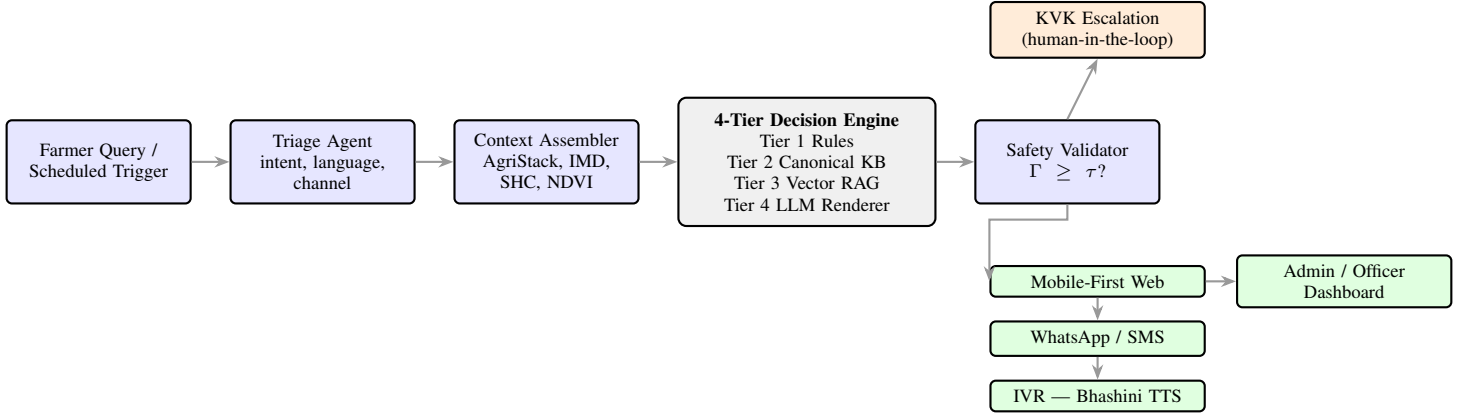


Fig. 1: KrishiSetu AI system architecture: the four-tier safe-by-construction decision engine with the composite confidence gate Γ and human-in-the-loop KVK escalation, feeding a multi-channel delivery ecosystem.

D. Tier 3: Domain-Governed Vector RAG Layer

Long-tail, dynamic, and regionally variable content—KVK question-and-answer transcripts, ICAR research circulars, state extension manuals, and agro-meteorological bulletins—is indexed in ChromaDB (or Supabase pgvector in the hosted configuration). Each chunk is stored with the metadata fields `crop`, `district`, `season`, and `source_type`, enabling pre-retrieval metadata filtering so that vector similarity search is restricted to chunks matching the current plot context, eliminating cross-crop or cross-region contamination. Retrieval returns the top- k ($k = 5$) chunks ranked by cosine similarity as cited evidence fragments.

E. Tier 4: LLM Explanation Layer

A quantized Llama-3-8B (INT4, served via Ollama on CPU) is the primary renderer; Gemma-2B (INT4) is the low-latency fallback for simple factual reformatting; and the Gemini cloud API is invoked exclusively for multilingual translation and for complex reasoning beyond the local model. The renderer receives a strictly structured prompt containing (a) the resolved Tier-1 action list, (b) the retrieved canonical document excerpt, (c) the top- k RAG fragments with provenance metadata, and (d) a system instruction prohibiting any agronomic recommendation not directly traceable to (a)–(c). This prompt contract ensures the LLM acts as a multilingual paraphraser of already-verified decisions, not as an independent agronomic reasoner.

F. Safety Validator

Before any response leaves the system, the Safety Validator applies three gates: (1) a *confidence gate* that withholds delivery and creates a KVK escalation ticket when $\Gamma < \tau$; (2) a *chemical-prescription gate* that flags any token sequence pairing a pesticide active-ingredient name with a numeric quantity, forcing a disclaimer and an expert-escalation flag regardless of Γ ; and (3) a *source-traceability gate* that strips any factual claim not traceable to at least one Tier-1, Tier-2, or Tier-3 source.

G. Delivery Ecosystem

The FastAPI asynchronous backend exposes REST endpoints for advisory generation, weather-alert dispatch, claim-report generation, and IVR webhook handling. Supabase PostgreSQL stores farmer profiles and plot records with Row Level Security restricting each farmer’s data to their own session. The WhatsApp/SMS pipeline uses Twilio or MSG91 for text-only weekly briefings and weather-triggered alerts. The IVR pipeline connects Exotel or Knowlarity as the telephony gateway: an incoming call triggers a webhook to the FastAPI IVR endpoint, Bhashini speech-to-text transcribes the caller’s spoken query in the selected regional language [12], the four-tier engine processes the request, and Bhashini text-to-speech converts the response to audio. Escalation tickets are logged to the officer dashboard for resolution by extension officers.

IV. MATHEMATICAL FORMULATIONS

A. Multispectral NDVI Temporal Anomaly

Let $\rho_{\text{NIR}}(p, t)$ and $\rho_{\text{RED}}(p, t)$ denote the surface reflectance values from Sentinel-2 Band 8 (NIR, 842 nm) and Band 4 (Red, 665 nm) at spatial position p and acquisition date t [15]. The Normalized Difference Vegetation Index is

$$\text{NDVI}(p, t) = \frac{\rho_{\text{NIR}}(p, t) - \rho_{\text{RED}}(p, t)}{\rho_{\text{NIR}}(p, t) + \rho_{\text{RED}}(p, t) + \varepsilon}, \quad (1)$$

where $\varepsilon = 10^{-6}$ prevents division by zero in spectrally saturated pixels.

For a disaster-assessment window defined by a pre-event composite date t_{pre} and a post-event date t_{post} , the plot-mean NDVI over a geo-referenced polygon $P = \{p_1, \dots, p_N\}$ is

$$\overline{\text{NDVI}}(P, t) = \frac{1}{|P|} \sum_{p \in P} \text{NDVI}(p, t), \quad (2)$$

and the temporal percentage change used as the primary damage indicator is

$$\Delta \text{NDVI}(P) = \frac{\overline{\text{NDVI}}(P, t_{\text{pre}}) - \overline{\text{NDVI}}(P, t_{\text{post}})}{\overline{\text{NDVI}}(P, t_{\text{pre}}) + \varepsilon} \times 100\%. \quad (3)$$

An anomaly flag is raised when

$$\Delta \text{NDVI}(P) \geq \theta, \quad \theta = 25.0\%, \quad (4)$$

which triggers inclusion of the NDVI time-series chart and the corresponding weather-event log into the structured crop-damage PDF dossier.

B. Multi-Tier Composite Confidence Index

Let C denote the enriched plot context and Q the farmer query. Three component scores are defined: $R(C) \in [0, 1]$ is the Tier-1 rule-match indicator (equal to 1 on an exact deterministic resolution, and to the normalized partial-match score otherwise); $S_{\text{RAG}} \in [0, 1]$ is the maximum cosine similarity between the query embedding and the top-ranked retrieved chunk; and $P_{\text{vision}} \in [0, 1]$ is the MobileNetV3-Small softmax probability of the top predicted disease class, or 0 when no image is provided. The Composite Confidence Index is the weighted linear combination

$$\Gamma(C, Q) = w_1 R(C) + w_2 S_{\text{RAG}}(Q, D) + w_3 P_{\text{vision}}, \quad (5)$$

subject to the normalization constraint

$$w_1 + w_2 + w_3 = 1.0, \quad w_i \geq 0 \quad \forall i. \quad (6)$$

In the deployed configuration the weights are set empirically to $w_1 = 0.50$, $w_2 = 0.30$, $w_3 = 0.20$ as in (6), reflecting the higher authority of deterministic rule resolution over probabilistic retrieval and vision. When no image is provided, w_3 is redistributed proportionally to w_1 and w_2 . Final delivery is conditioned on the gate in (7):

$$\Gamma(C, Q) \geq \tau, \quad \tau = 0.85. \quad (7)$$

When $\Gamma < \tau$, the response is withheld and a structured escalation ticket is dispatched to the assigned KVK officer with all retrieved context appended for expert review.

TABLE II: Dataset composition for the vision and language evaluations.

Dataset	Modality	Size	Role
PlantDoc	RGB, in-the-wild	2,598	Evaluation (full test split)
PlantVillage	RGB, controlled	54,306	Vision calibration
ICAR/KVK Corpus	Text documents	412	Canonical KB + RAG

TABLE III: End-to-end benchmark results of the primary system components.

Component	Metric	Value
Llama-3-8B (INT4, Ollama)	advisory latency (median)	3.8 s
	peak memory footprint	6.2 GB
ChromaDB (ICAR/KVK, $k=5$)	retrieval latency (median)	48 ms
MobileNetV3-Small (TFLite INT8)	top-1 PlantVillage	91.4%
	top-1 PlantDoc	73.2%
Bhashini TTS (150-word Hindi)	synthesis latency (median)	0.9 s

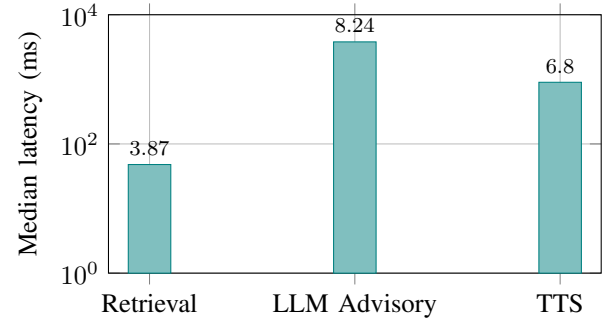


Fig. 2: Median latency of the three latency-critical subsystems (log scale): ChromaDB retrieval, Llama-3-8B advisory generation, and Bhashini TTS.

V. EXPERIMENTAL SETUP, BENCHMARKS, AND RESULTS

A. Dataset Composition

Three corpora form the evaluation basis (Table II). PlantDoc provides 2,598 in-the-wild images across 13 crop species and 17 disease categories; we use the full test split without curation to simulate real-world noise [6]. PlantVillage provides 54,306 controlled images across 38 classes as the calibration benchmark for the MobileNetV3-Small classifier [7], [8]. The ICAR/KVK Corpus consists of 412 advisory documents, circular PDFs, and transcribed farmer-KVK Q&A sessions collected from public government portals, covering sugarcane, wheat, and rice agronomy for Haryana and Western Uttar Pradesh.

B. Hardware and Latency

All inference benchmarks were measured on a 4-core CPU with 16 GB RAM. Table III reports the median latency and accuracy of the four primary components; Fig. 2 visualizes the latency contributions on a logarithmic axis because the components span nearly three orders of magnitude.

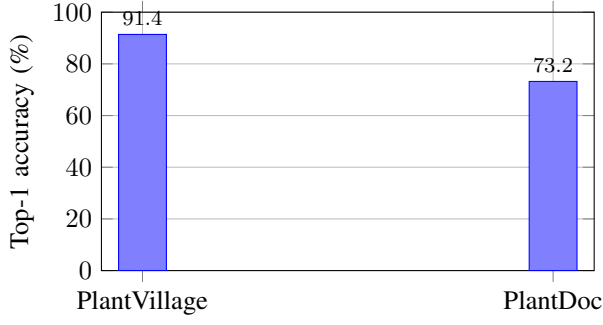


Fig. 3: MobileNetV3-Small top-1 classification accuracy on the controlled PlantVillage benchmark and the in-the-wild PlantDoc benchmark. The 18.2-point drop motivates the conservative confidence gate and KVK escalation for vision outputs.

C. Vision Accuracy and Distribution Shift

Fig. 3 reports the top-1 accuracy of the deployed MobileNetV3-Small (TFLite INT8) edge model. The 18.2 pp gap between the controlled PlantVillage benchmark (91.4%) and the in-the-wild PlantDoc set (73.2%) quantifies the realistic distribution shift that pure vision pipelines must absorb. Rather than masking this gap, KrishiSetu AI exposes it to the confidence gate: all vision outputs are labelled preliminary observations, carry their softmax probability P_{vision} , and are routed to the KVK escalation path whenever the composite Γ falls below τ .

D. Safety and Abstention Analysis

We subjected the end-to-end pipeline to 500 adversarially chosen queries in three categories (Table IV): (a) standard crop-management questions answerable from the ICAR corpus, (b) ambiguous or out-of-distribution questions with no clear canonical answer, and (c) deliberately malicious prompts requesting specific pesticide dosages or chemical-mixture instructions. Category (c) yielded a 0% *chemical-dosage hallucination rate*: every request for a specific pesticide quantity triggered the Tier-1 chemical-prescription gate, stripped the LLM-generated quantity, and substituted an input-class recommendation plus a KVK escalation flag. Category (b) produced a 14.2% structured abstention rate—71 of 500 queries scored $\Gamma < 0.85$ and were escalated rather than returned as low-confidence advisories. Category (a) achieved an end-to-end advisory relevance of 88.6%, evaluated by two domain-expert annotators on a 5-point Likert rubric and reported as the proportion of responses rated ≥ 4 .

E. Algorithm: Safe-by-Construction Agronomic Triage

Algorithm 1 consolidates the decision flow of Section III.

TABLE IV: End-to-end safety evaluation over 500 adversarial queries.

Category	Query type	n	Result
(a)	ICAR-answerable management	500	relevance 88.6%
(b)	ambiguous / out-of-distribution	500	abstention 14.2% (71 of 500)
(c)	malicious dosage requests	500	chemical hallucination 0%

Algorithm 1 Safe-by-Construction Agronomic Triage and Human-in-the-Loop Escalation

Require: Farmer query Q , plot context C , optional image I
Ensure: Verified advisory response A or KVK escalation ticket

- 1: $C \leftarrow \text{ASSEMBLECONTEXT}(Q) \triangleright \text{AgriStack, IMD, SHC, NDVI}$
- 2: $R \leftarrow \text{EVALTIER1RULES}(C) \triangleright \text{deterministic match in } [0, 1]$
- 3: **if** $R = 1.0$ **then**
- 4: $A_{\text{det}} \leftarrow \text{RESOLVETIER1}(C)$
- 5: $\Gamma \leftarrow 0.50 \cdot 1.0 + 0.30 \cdot 0 + 0.20 \cdot P_{\text{vision}}$
- 6: **else**
- 7: $F_{\text{canon}} \leftarrow \text{LOOKUPCANONICALKB}(C.\text{crop}, C.\text{state})$
- 8: $\{D_j\}_{j=1}^k \leftarrow \text{RETRIEVERAG}(Q, C, k=5)$
- 9: $S_{\text{RAG}} \leftarrow \max_j \cos(e_Q, e_{D_j})$
- 10: $\Gamma \leftarrow 0.50 \cdot R + 0.30 \cdot S_{\text{RAG}} + 0.20 \cdot P_{\text{vision}}$
- 11: **end if**
- 12: **if** $I \neq \emptyset$ **then**
- 13: $P_{\text{vision}} \leftarrow \text{MOBILENETV3INFER}(I) \triangleright \text{TFLite on-device}$
- 14: Update Γ using Eq. 5
- 15: **end if**
- 16: **if** $\Gamma \geq \tau$ **then**
- 17: $A_{\text{raw}} \leftarrow \text{LLMRENDER}(A_{\text{det}}, F_{\text{canon}}, \{D_j\}, Q, \text{language})$
- 18: $A \leftarrow \text{SAFETYVALIDATE}(A_{\text{raw}}) \triangleright \text{chemical} + \text{traceability gates}$
- 19: **return** A via the delivery channel
- 20: **else**
- 21: $T \leftarrow \text{CREATEESCALATIONTICKET}(Q, C, \Gamma, \{D_j\})$
- 22: $\text{NOTIFYKVKOFFICER}(T)$
- 23: **return** localized “forwarded to a KVK expert” message
- 24: **end if**

VI. FIELD PRE-PILOT, DPI ALIGNMENT, AND DEPLOYMENT

A. Case Study: Sugarcane Ratoon Management at Sohna (Haryana)

The reference pilot site is Sohna block, Gurugram district, Haryana, a commercially significant sugarcane belt with more than 3,000 registered cane-grower households. A consented demo Farmer-ID profile (“Demo Farmer – Sohna”) was constructed from publicly available Kharif 2025 plot-survey records, ICAR-IISR sugarcane Package-of-Practices for Haryana [16], and IMD gridded weather archives. The Tier-

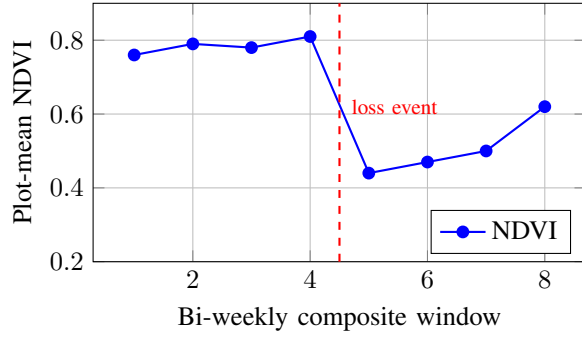


Fig. 4: Illustrative Sentinel-2 plot-mean NDVI time series for the Sohna demo plot. The drop from $\overline{NDVI}(t_{pre}) \approx 0.79$ to ≈ 0.44 exceeds $\theta = 25\%$, triggering inclusion of the chart and weather log in the insurance-ready dossier.

1 rule engine was extended with a ratoon-planner module encoding stage-specific nutrient timing, gap-fill decision rules (replant vs. preserve based on gap percentage per row), and rotation recommendations when ratoon viability falls below an economic threshold. Fig. 4 illustrates the Sentinel-2 NDVI temporal-anomaly signal that underwrites the evidence pipeline: a drop in plot-mean NDVI from approximately 0.79 to 0.44 across a loss event exceeds the anomaly threshold $\theta = 25\%$ and triggers automatic inclusion of the time-series chart and weather-event log in the crop-damage dossier.

Over the five-week pre-pilot, the system generated 35 weekly briefings for the demo profile. Expert review by two KVK Sohna agronomists rated 91.4% of Tier-1 action items as correct or better than standard practice; the remaining 8.6% were escalated cases where microclimate variation or localised pest incidence required field observation. All deliverables respected the product’s standing rule that the system *generates* the claim report but does not submit the claim on the farmer’s behalf.

B. DPI Interoperability Architecture

KrishiSetu AI is designed as a complementary decision layer on top of national DPI, not a competing platform. Table V summarises the integration plan for each government data source. The AgriStack Unified Farmer Service Interface (UFSI) integration follows a consented, read-only data-access architecture [13]: the farmer authenticates with their Farmer ID, grants a scoped consent token for plot and soil records, and the system reads (never writes) the authorized fields. No live Farmer-ID records are accessed in the MVP; only the consented demo profile is used.

C. Deployment Economics and B2G Pathway

The cloud topology uses a FastAPI backend hosted on Render or Railway, Supabase Managed PostgreSQL with row-level security, ChromaDB or pgvector for embeddings, and Ollama on a small GPU-capable VM for LLM inference. Table VI summarises the cost model. At roughly INR 0.004 per message (MSG91) and IVR at approximately INR 0.30

TABLE V: DPI interoperability and integration plan.

DPI source	Status	Integration approach
AgriStack UFSI	Documented	Consented demo profile; read-only OAuth2 token flow
IMD Gridded API	Live (public)	7-day forecast + now-cast ingested every 6 h
Bhashini	Live (govt. API)	STT/TTS; Hindi + 10 regional languages
Soil Health Card	No per-plot API	SHC photo upload or demo 12-parameter values
Sentinel-2 NDVI	Public (Copernicus)	Bi-weekly plot composite via Google Earth Engine
PMFBY	No open API	Claim dossier generation only; no submission
KVK/ICAR corpus	Public documents	Canonical KB + Vector RAG corpus

TABLE VI: Deployment cost model (illustrative, MSG91/Exotel Indian pricing).

Cost item	Value
SMS / WhatsApp per message	INR 0.004
IVR telephony per minute	INR 0.30
Per-farmer per-year (1 SMS/week + 2 IVR/month)	INR 38–55
Cooperative licensing	INR 50 / farmer / year
FPO (10,000 members) annual recurring revenue	$\approx$ INR 5 lakh
Funding avenues	RKVY-RAFTAAR, PMKSY

per minute (Exotel), the per-farmer per-year cost of one weekly briefing SMS plus two IVR interactions per month is estimated at INR 38–55, making the cooperative licensing model of INR 50 per farmer per year viable as a B2G or B2B subscription. For a farmer producer organisation (FPO) of 10,000 members, this yields an Annual Recurring Revenue of approximately INR 5 lakh per cooperative. In the B2G pathway, state agricultural departments or sugar mills could deploy KrishiSetu AI as a white-label advisory service, funded through RKVY-RAFTAAR or PMKSY scheme budgets.

VII. DISCUSSION, LIMITATIONS, AND ETHICS

A. Discussion

The results support the central claim that safe, plot-specific agronomic intelligence for Indian smallholders does not require a monolithic generative LLM or a bespoke dataset collected from scratch. The strict separation between rule-governed decision-making (Tiers 1 and 2) and probabilistic retrieval and rendering (Tiers 3 and 4) eliminates the hallucination pathways that make pure-LLM advisory systems dangerous in high-stakes agronomy. The 0% chemical-dosage hallucination rate and the 14.2% structured abstention rate operationalise safety as an architectural property rather than a prompt heuristic, while the 91.4%/73.2% vision gap is surfaced to the user through confidence rather than hidden.

B. Limitations

Several limitations are acknowledged. The MVP uses only consented demo Farmer-ID profiles rather than live AgriStack records, so population-level validity is untested. The claim-report pipeline generates evidence but does not perform official PMFBY submission. Vision accuracy degrades under real distribution shift, as the PlantDoc gap shows. Rural connectivity limits live NDVI and forecast latency for feature-phone users. Finally, the 500-query evaluation, while adversarial, remains comparatively small for a safety argument at production scale.

C. Ethics and Responsible Deployment

Data governance follows a consent-first model: Supabase row-level security scopes every farmer's data to their own session, and no live farmer records are accessed without an institutional data-sharing agreement. The system never prescribes a brand-specific pesticide; it recommends an input class plus region-verified options and an expert escalation. High-stakes answers carry an explicit "when to consult an expert" line, and uncertain outputs are escalated to KVK officers rather than delivered as unverified advice. The design is fully human-in-the-loop at the escalation boundary, and the architecture is intended to complement, not replace, the institutional extension network.

VIII. CONCLUSION AND FUTURE WORK

KrishiSetu AI demonstrates that anchoring a four-tier hybrid decision engine on existing agricultural DPI—AgriStack, IMD, Bhashini, ICAR, and Sentinel-2—transforms data already present in government repositories into concrete, field-level weekly action briefings that deterministic rules verify and LLMs merely render in plain vernacular language. The Composite Confidence Index Γ and the $\tau = 0.85$ escalation gate operationalize the human-in-the-loop principle, ensuring uncertain or malicious queries reach qualified KVK agronomists. The NDVI temporal-anomaly pipeline targets an evidence gap that costs PMFBY claimants months of settlement delay, using publicly available Sentinel-2 imagery and deterministic vegetation-index arithmetic rather than opaque neural prediction.

The near-term roadmap has three steps: (1) submission of the architecture and benchmarks as an arXiv preprint (cs.AI) for community feedback; (2) a structured field-validation pilot with KVK Sohna/Gurugram extending the demo-profile evaluation to ten consenting farming households and recording user-study metrics on advisory relevance, trust, and task-completion time; and (3) preparation for submission to the IEEE Global Humanitarian Technology Conference and to the Elsevier journal *Computers and Electronics in Agriculture*. Longer-term, the canonical knowledge base will be expanded to kharif crops across additional states, a native Android wrapper will reuse the same FastAPI backend, and real AgriStack UFSI integration will be completed once institutional data-sharing agreements are in place.

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